

★ UNIT-4 ⇒

• Data Analytics:

- The IoT is an network of interconnected devices and gadgets that can collect and share data by themselves.
- The data analytics refers to gathering ~~the~~ examining and data produced by these devices and make wise decisions.

Imp. Components:

- 1) Data collection.
- 2) Data storage.
- 3) Data Processing.
- 4) Data Analysing.
- 5) Data visualization.

Imp. Types of data analytics ⇒

- 1) Descriptive Analytics ⇒ It is the process of using current and historical data to identify trends and relationship.
E.g. ⇒ Financial statement analysis, trends analysis, tracking process etc.
- 2) Diagnostic Analysis ^{finds} / Root cause ⇒ It is the process of using data to determine the causes of trends and correlation between variables.

3) Predictive Analysis $\Rightarrow$ It uses the data to predict future trends and events.

E.g. $\Rightarrow$ Finance, marketing, manufacturing, health-care etc.

4) Prescriptive Analysis $\Rightarrow$

challenges $\Rightarrow$

- 1) Data security.
- 2) Data Privacy
- 3) Data Quality
- 4) Scalability.
- 5) Interoperability.

It uses both descriptive and predictive analytics and gives the feel of diagnostic analysis.

• 5 V's

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|---------------------|-------------|
| volume | 1) volume |
| velocity | 2) velocity |
| variety | 3) variety |
| | 4) veracity |
| | 5) value |